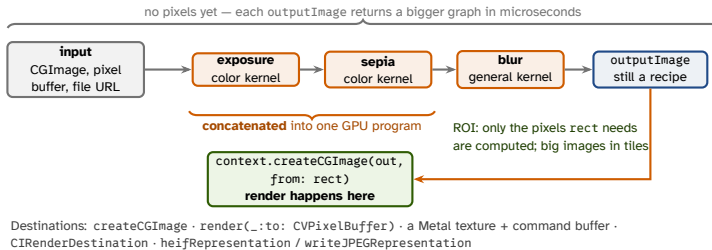
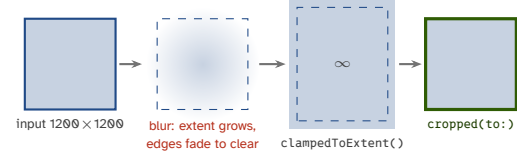


In one line: A CIImage is a **recipe, not pixels**: applying a CIFilter only extends a lazy graph. Nothing is computed until a CIContext **renders** a rectangle of it — then Core Image walks the graph *backwards* from that rectangle (region of interest), **concatenates** adjacent kernels into as few GPU (Metal) passes as it can, and writes the result to a CGImage, pixel buffer, texture or file. Contexts are **expensive** — **make one and reuse it**; contexts and images are immutable and thread-safe, filters are mutable and are not.

A. Build a recipe (free) → render a rectangle (all the work)



B. Extent — the blur trap



Outside its extent an image is clear black. Clamp (repeat the edge pixels, infinite extent) before the blur, crop back to input.extent after. Generators (CIColorGenerator, checkerboard) are infinite too: rendering .infinite fails — always crop.

How it works

- Inputs:** CIImage(cgImage:), (cvPixelBuffer:) (camera frames, no copy), (image:) from UIImage — which drops imageOrientation: apply .oriented(_:). Origin is **bottom-left**, like Quartz.
- Two filter APIs:** stringly CIFilter(name: "CIGaussianBlur") + setValue(_:forKey: kCIInputImageKey) (typos = runtime nil), or typed import CoreImage.CIFilterBuiltins (iOS 13) → CIFilter.gaussianBlur() with inputImage / radius properties.
- Lazy + concatenated:** chains of per-pixel (colour) kernels fuse into one program — no intermediate texture per step; a blur or warp needs neighbours, so it may force one.
- CIContext** owns an MTLCommandQueue, the compiled-kernel cache and intermediate buffers. Apple: *one per view that renders, or per background task* — never per image or per frame. CIContext(mtlDevice:) / (mtlCommandQueue:) shares your Metal pipeline; .useSoftwareRenderer: true forces the (slow) CPU path.
- Colour:** inputs are converted into the **working space** — default **extended sRGB, linear gamma**, so maths happens in linear light and values beyond 0–1 survive; output converts to the destination's space. Masks, depth, data: .workingColorSpace: NSNull() turns colour management off.
- Custom kernels** are Metal (MSL): CIColorKernel (one pixel in, one out), CIWarpKernel (returns the *source coordinate*), CIBlendKernel, general CIKernel (samplers + an ROI callback). Build: *Other Metal Compiler Flags* -fcikernel, user setting MTLINKER_FLAGS = -cikernel. String kernels (CIKernel(source:)) are **deprecated since iOS 12**. MPS or a model inside the graph: CIImageProcessorKernel.
- Vision hand-off:** the same CIImage feeds VNImageRequestHandler(ciImage:orientation:); results are **normalised** (0–1), **bottom-left** origin — convert with VNImageRectForNormalizedRect, then e.g. pixellate only the face rect and composite it back.

Example — one shared context, typed filters

```
import CoreImage.CIFilterBuiltins
let context = CIContext() // ONE, app-wide or per view
func stylise(_ cg: CGImage) -> CGImage? {
    let input = CIImage(cgImage: cg)
    let blur = CIFilter.gaussianBlur() // typed API, iOS 13
    blur.inputImage = input.clampedToExtent() // no fading edges
    blur.radius = 8
    let tone = CIFilter.sepiaTone() // filters: NOT thread-safe
    tone.inputImage = blur.outputImage; tone.intensity = 0.6
    guard let out = tone.outputImage?.cropped(to: input.extent)
    else { return nil } // still only a recipe
    return context.createCGImage(out, from: out.extent) // GPU work: HERE
}
```

Custom colour kernel (MSL) + applying it

```
#include <CoreImage/CoreImage.h> // Threshold.ci.metal
extern "C" float4 threshold(coreimage::sample_t s, float t) {
    float l = dot(s.rgb, float3(0.2126, 0.7152, 0.0722)); // luma
    return float4(float3(step(t, l)), s.a); // black or white
} // Swift side:
let k = try CIColorKernel(functionName: "threshold",
                           fromMetalLibraryData: metalLibData)
let bw = k.apply(extent: img.extent, arguments: [img, 0.5])
```

Live camera — stay on the GPU

AVCaptureVideoDataOutput → CVPixelBuffer → CIImage(cvPixelBuffer:) → filters → context.render into the MTKView drawable's texture (or a CIORenderDestination) — never createCGImage per frame: that is a GPU→CPU readback plus an allocation, 30–60 times a second.

Which tool for an image effect?

Core Image	built-ins, graph optimisation, photo + video, RAW
SwiftUI shader	one view's pixels (.colorEffect / .layerEffect, iOS 17)
Metal	full control: your own passes, compute, 3D, ML
Core Graphics	drawing shapes and text, not filtering pixels

Traps

- ! A new CIContext per image or per frame — kernel recompiles, memory spikes, stutter.
- ! UIImage(ciImage:) in a UIImageView: still a recipe, rendered on every draw by a context you do not control. Render to a CGImage with your shared context.
- ! Soft transparent borders after a blur, or createCGImage(out, from: out.extent) is nil (infinite extent) — clamp, then crop.
- ! Rotated photo — CIImage(image:) drops the orientation.
- ! One CIFilter shared between threads — a data race; share images and the context instead.
- ! Face box drawn in the wrong place — Vision is normalised and bottom-left, UIKit top-left.

Remember

"CIImage is the recipe, CIContext is the kitchen — build the kitchen once, cook only what's ordered."